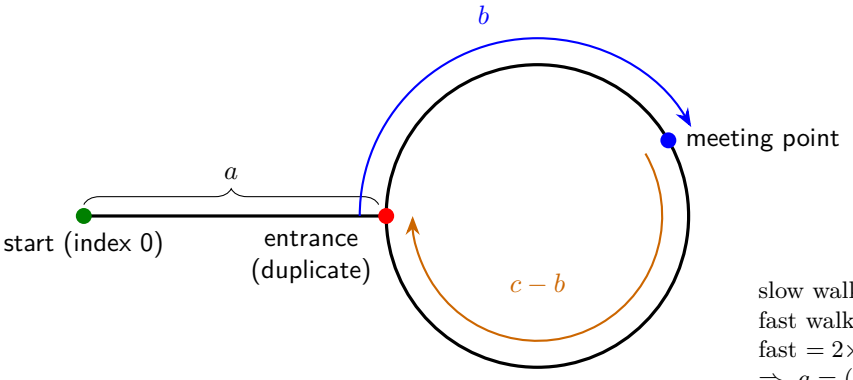

slow walked $a + b$
fast walked $a + b + k \cdot c$
fast $= 2\times$ slow $\Rightarrow\ a + b = k \cdot c$
$\Rightarrow\ a = (k-1)\,c + (c-b)$